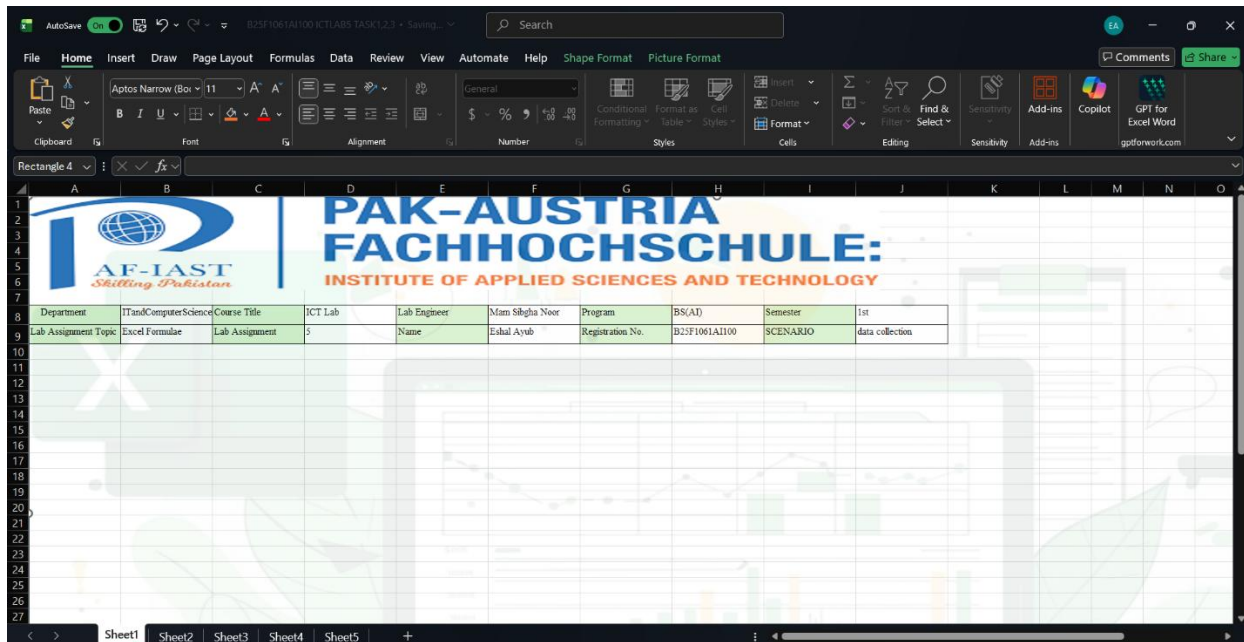




PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

ICT Lab 05	Instructor: Ms,Quratulain
Program: AI- Fall 2025	Date: 9th November, 2025
Marks: 10	Lab Engineer: Ms. Sibgha Noor
Student Name: Eshal Ayub	Reg. No: B25F1061AI100
Instructions: • Submit the file with your names following your registration numbers like AI001_Name. • Copy Paste work will be given no marks.	

Introduction:



Lab Learning Objectives

The primary objectives of this lab were to:

1. Learn how to use Excel formulas such as **SUM**, **AVERAGE**, and **IF** to calculate results efficiently.
 2. Understand how to design a **CGPA Calculation System** using weighted averages.
 3. Explore the use of **PivotTables** and **Pivot Charts** to summarize and visualize data.
 4. Develop practical data presentation skills, including using **Map Charts** and conditional functions.
 5. Apply formatting and layout techniques for professional and accurate data representation.
-

Lab Tasks

Task 1: CGPA Calculation System

Objective:

To create an Excel system that calculates Semester GPA and CGPA for a dataset of students using Excel formulas.

Procedure:

1. Created a dataset starting from **row 4** with columns: *Name, Roll No, Sub1, Sub2, Sub3, Credit Hours, Grade Points, GPA, CGPA*.
2. Entered data for 15 students, assigning random marks and grade points.
3. Used the formula `=AVERAGE (C4:E4)` in column H to calculate **Semester GPA**.
4. Added a formula `=AVERAGE (H4:H18)` to calculate the **overall CGPA**.
5. Applied cell formatting shows results up to **two decimal places**.
6. Used borders, table formatting, and alignment for neat presentation.

Screenshot:

Name	Roll No	Sub1	Sub2	Sub3	Credit Hours	Grade Points	GPA	CGPA
Adeel	1	78	85	90	3	4	84.33	3.9
Sana	2	65	70	68	3	3.2	67.67	3.2
Zoya	3	55	60	72	3	2.8	62.33	2.8
Bilal	4	88	92	85	3	3.9	88.33	3.9
Hira	5	74	80	76	3	3.3	76.67	3.3
Daniyal	6	49	58	65	3	2.5	57.33	2.5
Mahnoor	7	90	87	93	3	4	90	4
Ahmed	8	68	72	70	3	3	70	3
Zara	9	81	79	84	3	3.7	81.33	3.7
Hassan	10	59	63	61	3	2.6	61	2.6
Areeba	11	77	82	79	3	3.5	79.33	3.5
Rehan	12	84	88	91	3	3.9	87.67	3.9
Nida	13	69	74	73	3	3.1	72	3.1
Farhan	14	61	66	70	3	2.9	65.67	2.9
Iqra	15	92	95	94	3	4	4	4

Task 2: PivotTable & PivotChart

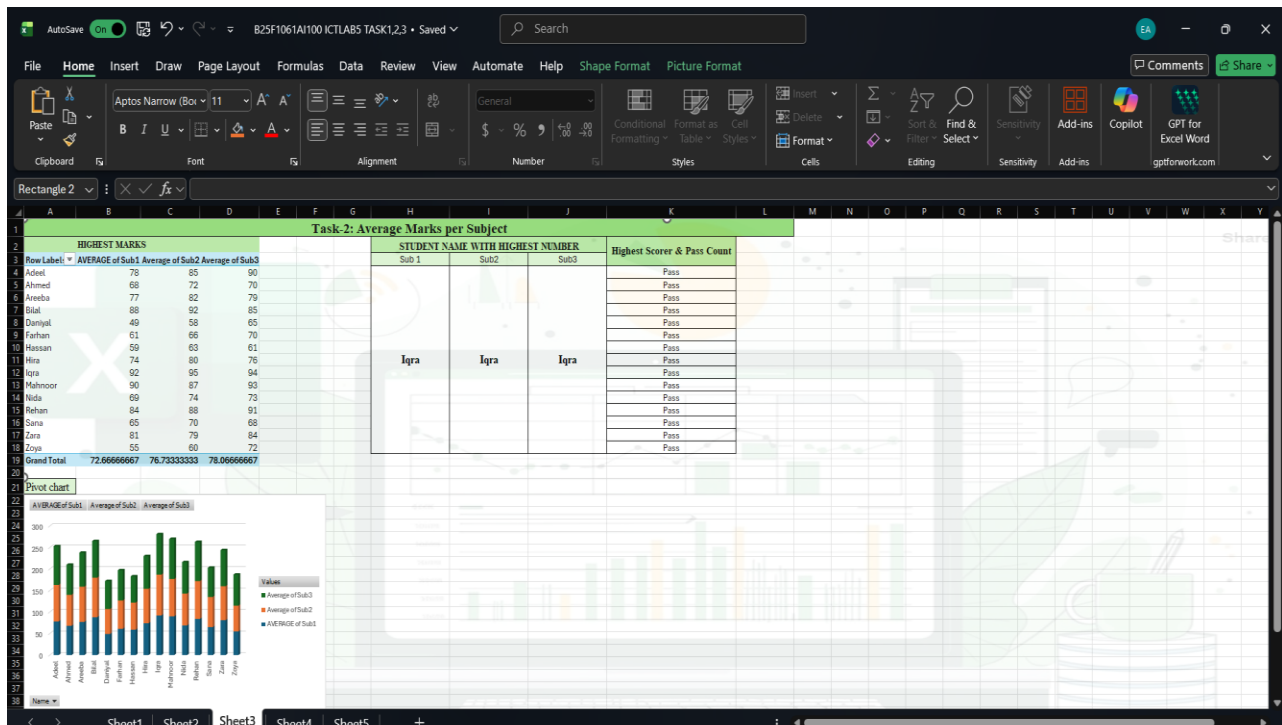
Objective:

To summarize student performance and visualize the data using PivotTable and PivotChart tools.

Procedure:

1. Selected the dataset from Task 1 and inserted a **PivotTable** (Insert → PivotTable).
2. Dragged **Student Name** to *Rows* and **Subjects (Sub1, Sub2, Sub3)** to *Values*.
3. Changed the Value Field Setting to **Average** to display average marks per subject.
4. Inserted a calculated column with `=IF(AVERAGE (C4:E4)>=50,"Pass","Fail")` to count passing students.
5. Created a **PivotChart** (Column type) to visualize average marks for each subject.
6. Adjusted the chart design and added data labels for clarity.

Screenshot:



Task 3: Country Data and Map Visualization

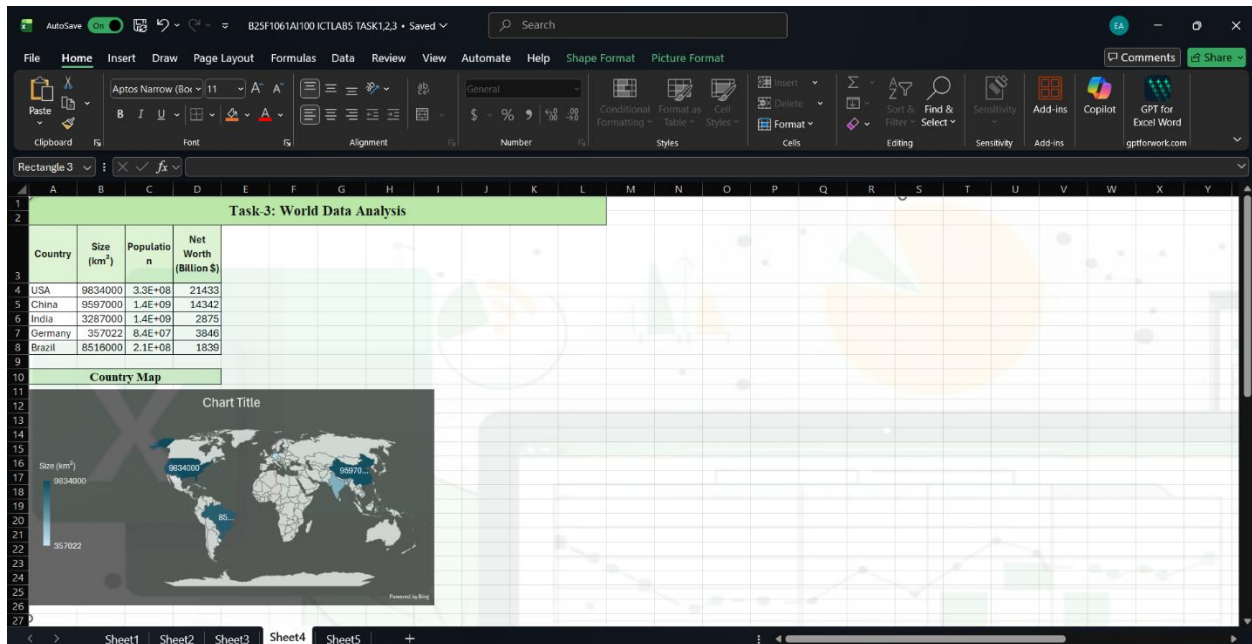
Objective:

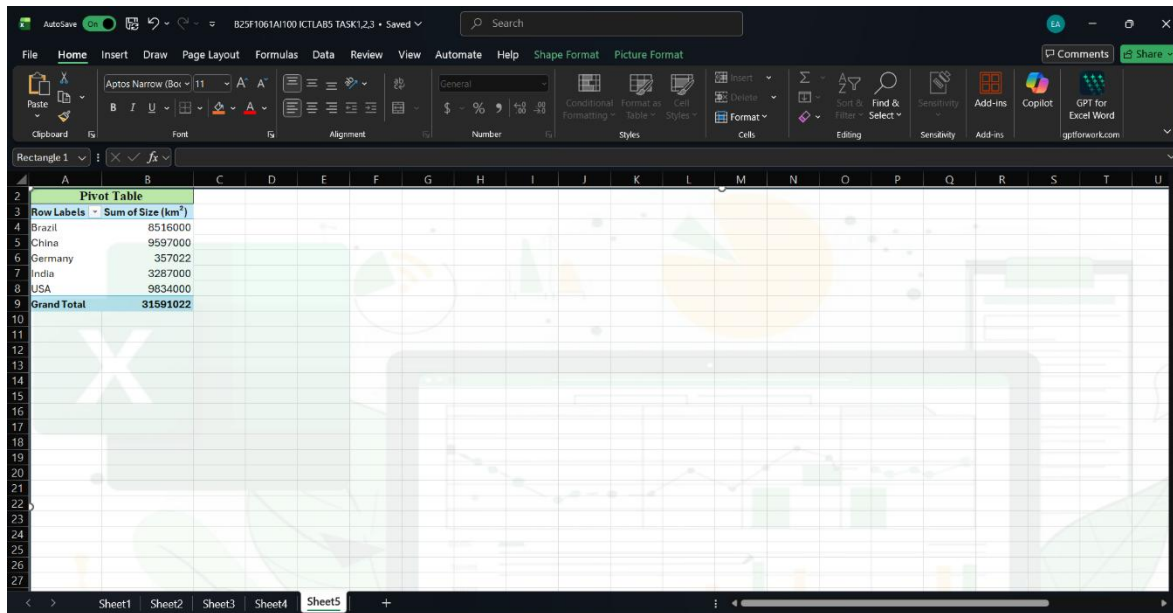
To create a dataset of countries and visualize it using a World Map chart, PivotTable, and logical Excel functions.

Procedure:

1. Created a new sheet with columns: *Country*, *Size*, *Population*, *Net Worth*, *Population/Size*.
2. Entered data for 5 countries (Pakistan, USA, Japan, France, Canada).
3. In the last column, the formula $=C4/B4$ entered to calculate **Population** ÷ **Size**.
4. Inserted a **PivotTable** with *Country* and *Size* columns.
5. Inserted a **World Map Chart** (Insert → Maps → Filled Map) to visualize population distribution.
6. Added a conditional formula in an empty cell:
7. $=IF(C4<C5,"Country has lower population", "Country has higher population")$
8. Inserted a background image for better presentation (Page Layout → Background).

Screenshot:





Summary:

This lab improved my understanding of how Excel can be used not just for basic calculations, but for **data analysis, visualization, and automation**.

I learned how to use formulas, create PivotTables for summary reports, and visualize geographic data using Map Charts.

Overall, the lab strengthened my skills in **data organization, analytical thinking, and report formatting**, which are essential for professional ICT work.

